

## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

### ASSESSMENT TOOL 2 – Pseudocode Writing Task

|                  |                                                                                      |               |                                             |
|------------------|--------------------------------------------------------------------------------------|---------------|---------------------------------------------|
| Course Code:     | CSA06                                                                                | Course Title: | Fundamentals of Data science                |
| CO Assessed:     | CO5 – Naturalization (BL5, BL6)                                                      | Assessment:   | Assessment Tool 2 – Pseudocode Writing Task |
| Weightage:       | 30% of CIA                                                                           | Total Marks:  | 100                                         |
| CO5 Statement:   | To understand the concept of supervised and unsupervised methods in machine learning |               |                                             |
| Student Name:    | U.Midhula                                                                            | Reg. No.:     | 192472115                                   |
| Section / Batch: | Cse-ai                                                                               | Date:         | 24-8-26                                     |

#### Instructions to Students

- Answer the following question with proper pseudocode and logical explanation.
- Clearly specify the algorithmic approach and complexity class wherever applicable.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) for complexity analysis.
- Include state-space representation, recursion steps, and pruning conditions in backtracking problems.
- Neat presentation and step-by-step justification are mandatory.

| Question Type           | Focus Area                                                                           | Questions |
|-------------------------|--------------------------------------------------------------------------------------|-----------|
| Pseudocode Writing Task | To understand the concept of supervised and unsupervised methods in machine learning | 5         |

#### SECTION A – Pseudocode Writing Task

##### Code of Conduct

I 192472115(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:U.Midhula

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                                   | Level 3 – Proficient                      | Level 2 – Developing         | Level 1 – Beginning             |
|---------------------------|-----------|-----------------------------------------------------------------------|-------------------------------------------|------------------------------|---------------------------------|
| Problem Understanding     | 20%       | Identifies all inputs, outputs, constraints accurately                | Minor missing elements                    | Partial understanding        | Incorrect interpretation        |
| Algorithm Design          | 30%       | Complete, correct, and logically sound solution                       | Minor logical gaps                        | Partially correct logic      | Incorrect approach              |
| Use of Control Structures | 20%       | Efficient and appropriate use of loops, conditions, functions/modules | Minor inefficiencies                      | Limited or basic usage       | Incorrect or missing constructs |
| Pseudocode Structure      | 10%       | Well-structured, clear, consistent format, easy to follow             | Mostly clear with minor issues            | Difficult to follow in parts | Poorly structured and unclear   |
| Completeness              | 20%       | Handles all cases; optimized approach                                 | Handles most cases; reasonable efficiency | Handles basic cases only     | Incomplete or inefficient       |

| Criteria                  | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing |
|---------------------------|-----|---------------------|----------------------|----------------------|
| Problem Understanding     | 20% |                     |                      |                      |
| Algorithm Design          | 30% |                     |                      |                      |
| Use of Control Structures | 20% |                     |                      |                      |
| Pseudocode Structure      | 10% |                     |                      |                      |
| Completeness              | 20% |                     |                      |                      |
| <b>Total Marks (100):</b> |     |                     |                      |                      |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
| Name:             | Name:              | Name:              |
| Signature: _____  | Signature: _____   | Signature: _____   |

1. Design pseudocode to train a linear regression model, calculate predicted values and residuals, and determine the **Mean Squared Error (MSE)** and **R<sup>2</sup> score**.

**Problem Statement:**

Design an algorithm to train a Linear Regression model, calculate predicted values and residuals, and evaluate the model using Mean Squared Error (MSE) and R<sup>2</sup> score.

**Algorithm:**

1. Start.
2. Read the input features X and target values Y.
3. Train a Linear Regression model using X and Y.
4. Calculate predicted values Y\_pred.
5. Calculate residuals: Residual = Y - Y\_pred.
6. Calculate MSE.
7. Calculate R<sup>2</sup> score.
8. Display predictions, residuals, MSE, and R<sup>2</sup> score.
9. Stop.

**Pseudocode:**

BEGIN

INPUT X, Y

Train Linear Regression model using X and Y

Y\_pred ← Predict values using trained model

FOR each data point

```

    Residual ← Y - Y_pred
END FOR

MSE ← Sum((Y - Y_pred)2) / Number of data points

R2 ← 1 - [Sum((Y - Y_pred)2) / Sum((Y - Mean(Y))2)]

PRINT Y_pred
PRINT Residual
PRINT MSE
PRINT R2

END

```

### Sample Output:

```

Predicted Values: [50, 60, 70, 80]
Residuals:        [2, -1, 1, -2]
MSE:              2.50
R2 Score:        0.96

```

2.Design pseudocode for **logistic regression** to predict whether a student will pass or fail based on study hours and attendance percentage.

### Problem Statement:

Design an algorithm using Logistic Regression to predict whether a student will **Pass or Fail** based on study hours and attendance percentage.

### Algorithm:

1. Start.
2. Read study hours and attendance percentage.

3. Read corresponding pass/fail results.
4. Train a Logistic Regression model.
5. Give a new student's study hours and attendance.
6. Predict the class.
7. Display Pass or Fail.
8. Stop.

**Pseudocode:**

BEGIN

INPUT study\_hours, attendance

INPUT result

Train Logistic Regression model using  
study\_hours and attendance

INPUT new\_study\_hours

INPUT new\_attendance

prediction ← Model\_Predict(new\_study\_hours, new\_attendance)

IF prediction = 1 THEN

    PRINT "PASS"

ELSE

    PRINT "FAIL"

END IF

END

**Sample Input:**

Study Hours = 6

Attendance = 85%

**Sample Output:**

Prediction: PASS

3.Design pseudocode that calculates the sigmoid probability for logistic regression and converts the probability into a binary class using a threshold of 0.5

**Problem Statement:**

Design an algorithm to calculate the sigmoid probability in Logistic Regression and convert the probability into a binary class using a threshold of 0.5.

**Algorithm:**

1. Start.
2. Take the linear value  $z$ .
3. Calculate sigmoid probability.
4. Compare probability with 0.5.
5. If probability  $\geq 0.5$ , assign class 1.
6. Otherwise, assign class 0.
7. Display probability and class.
8. Stop.

**Pseudocode:**

BEGIN

INPUT  $z$

Probability  $\leftarrow 1 / (1 + e^{(-z)})$

IF Probability  $\geq$  0.5 THEN

    Class  $\leftarrow$  1

ELSE

    Class  $\leftarrow$  0

END IF

PRINT Probability

PRINT Class

END

**Sample Input:**

$z = 2$

**Sample Output:**

Sigmoid Probability = 0.8808

Binary Class = 1

14. Design pseudocode to evaluate a logistic regression classifier using the **confusion matrix, accuracy, precision, recall, and F1-score**

**Problem Statement:**

Design an algorithm to evaluate a Logistic Regression classifier using a confusion matrix, accuracy, precision, recall, and F1-score.

**Algorithm:**

1. Start.
2. Read actual and predicted class values.
3. Calculate TP, TN, FP, and FN.

4. Construct the confusion matrix.
5. Calculate accuracy.
6. Calculate precision.
7. Calculate recall.
8. Calculate F1-score.
9. Display all evaluation measures.
10. Stop.

**Pseudocode:**

BEGIN

INPUT Actual\_Values

INPUT Predicted\_Values

TP  $\leftarrow$  0

TN  $\leftarrow$  0

FP  $\leftarrow$  0

FN  $\leftarrow$  0

FOR each data point

IF Actual = 1 AND Predicted = 1 THEN

TP  $\leftarrow$  TP + 1

ELSE IF Actual = 0 AND Predicted = 0 THEN

TN  $\leftarrow$  TN + 1

ELSE IF Actual = 0 AND Predicted = 1 THEN

FP  $\leftarrow$  FP + 1



```

ELSE
    FN  $\leftarrow$  FN + 1
END IF

END FOR

Accuracy  $\leftarrow$  (TP + TN) / (TP + TN + FP + FN)

Precision  $\leftarrow$  TP / (TP + FP)

Recall  $\leftarrow$  TP / (TP + FN)

F1_Score  $\leftarrow$  2  $\times$  Precision  $\times$  Recall / (Precision + Recall)

PRINT Confusion Matrix
PRINT Accuracy
PRINT Precision
PRINT Recall
PRINT F1_Score

END

```

**Sample Input:**

Actual: 1 1 0 0 1 0

Predicted: 1 0 0 1 1 0

**Sample Output:**

TP = 2

TN = 2

FP = 1

FN = 1

Confusion Matrix:

|          | Predicted |   |
|----------|-----------|---|
|          | 0         | 1 |
| Actual 0 | 2         | 1 |
| 1        | 1         | 2 |

Accuracy = 0.67

Precision = 0.67

Recall = 0.67

F1-Score = 0.67

5.Design pseudocode for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

**Problem Statement:**

Design an algorithm for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups using clustering, and produces cluster assignments.

**Algorithm:**

1. Start.
2. Read the unlabelled dataset.
3. Select relevant features.
4. Handle missing values.
5. Normalize or standardize the features.
6. Select the number of clusters.
7. Apply a clustering algorithm such as K-Means.
8. Assign each data point to a cluster.
9. Display the cluster assignments.

10. Stop.

**Pseudocode:**

BEGIN

INPUT Unlabelled\_Dataset

Select relevant features

Handle missing values

Normalize features

INPUT Number\_of\_Clusters K

Initialize K cluster centers

REPEAT

    Assign each data point to the nearest cluster center

    Calculate new cluster centers

UNTIL cluster assignments do not change

FOR each data point

    Assign Cluster\_Number

END FOR

PRINT Cluster Assignments

END

**Sample Input:**

Dataset:

| Study Hours | Attendance |
|-------------|------------|
| 2           | 50         |
| 3           | 55         |
| 4           | 60         |
| 7           | 80         |
| 8           | 85         |
| 9           | 90         |

Number of Clusters = 2

**Sample Output:**

Cluster 1:

(2,50)

(3,55)

(4,60)

Cluster 2:

(7,80)

(8,85)

(9,90)

Cluster Assignments:

1 1 1 2 2 2